

Chien-Hsien Chu

FULL-STACK • DEVOPS • PLATFORM ENGINEER • 5+ YEARS WORK EXPERIENCE

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Education

National Yang Ming Chiao Tung University (NYCU)

Hsinchu, Taiwan

M.S. IN COMPUTER SCIENCE

Sep. 2023 – Sep. 2026

- Overall GPA: 4.2/4.3 · Advisor: Prof. Chien-Chao Tseng
- Thesis: *Observability-Driven Root-Cause Analysis and Recovery Framework for Cloud-Native 5G Core Network Functions.*

Academic & Technical Experience

Observability-Driven Root-Cause Analysis and Recovery Framework for Cloud-Native 5G Core Network Functions

M.S. Thesis

GRADUATE RESEARCH ASSISTANT, WIRELESS & INTERNET LAB (WINLAB), NYCU

Feb. 2026 – Sep. 2026

- End-to-End Observability, RCA, and Recovery Framework for Multi-Cluster **Kubernetes**: Non-Intrusive Telemetry Collection, Generation, and Correlation, Fault Detection, Automated RCA, and Guarded Remediation.
- Telemetry Collection, Generation, and Correlation: **Zero-Code** Trace and RED Metric Generation via **eBPF (Grafana Beyla)**, shipped over **OTLP** by **OpenTelemetry Collectors**.
- Telemetry Storage and Fault Detection: **Prometheus, Loki, Tempo, Alertmanager**.
- LLM-Based RCA Agent: Agentic Investigation Loop over live telemetry, guided by injected environment knowledge.
- RAG-Based Recovery Agent: Knowledge Base retrieval, then **Auto / Approval / Escalation** remediation over **MCP**.
- Transparent State-Recovery Proxy: record-and-replay recovery of a stateful service's in-memory state after a restart.

🔗 Infrastructure Automation & Management for Laboratory Environment

🔗 [source code](#)

SYSTEM ADMINISTRATOR, WIRELESS & INTERNET LAB (WINLAB), NYCU

Jul. 2024 – Sep. 2026

- Independently migrated a Kubernetes platform from **Kubespray + Terraform** to an immutable, GitOps Architecture using **Talos Linux** and **Flux CD**, including cluster bootstrap, node lifecycle automation, and PV-level data migration.
- Executed multi-application, multi-version upgrades involving rollbackable database upgrades and data migrations.
- Integrated **Velero, MinIO, Grafana Stack**, and **Keycloak** for backup, object storage, observability, and access control.
- Managed **Harvester** clusters and migrated to **Proxmox VE** for improved stability and flexibility.

Kubernetes Networking and Security Enhancement for Department Infrastructure

SYSTEM TEACHING ASSISTANT, CSIT, NYCU

Jul. 2024 – Sep. 2026

- Migrated cluster networking from **Flannel** to **Calico**, enforcing fine-grained **NetworkPolicy** traffic isolation.
- Deployed **Istio Egress Gateway** to centralize outbound routing, policy enforcement, and auditability.
- Evaluated **External Secrets Operator (ESO)** and **OpenBao (Vault fork)** for cluster-wide secrets management.

Multi-Cluster Management System

GRADUATE STUDENT ASSISTANT, WIRELESS & INTERNET LAB (WINLAB), NYCU

Sep. 2023 – Jan. 2024

- **Multi-Cluster Management System** integrating **Karmada, Cilium**, and **Thanos** for orchestration and observability.

Kubernetes Infrastructure and DevOps Workflow Development

TECHNICAL ASSISTANT, INFORMATION TECHNOLOGY SERVICE CENTER, NYCU

Sep. 2023 – Aug. 2024

- Built scalable **Kubernetes** infrastructure for institutional systems.
- Enhanced CI/CD automation using **GitLab CI** to improve delivery reliability.

System Architecture Planning and Azure Infrastructure Consultation

TECHNICAL CONSULTANT, KOI STUDIO LTD.

Dec. 2023 – Nov. 2024

- Advised on system architecture and **Azure** infrastructure design for a software product company.
- Resolved deployment and integration issues between cloud services and external vendor APIs.

Work Experience

ISCOM Online International Information Inc.

Taichung, Taiwan

R&D ENGINEER, R&D DEPARTMENT

Jul. 2022 – Aug. 2023

- Built and operated production **Kubernetes** clusters across on-premises and cloud environments (**AKS**).
- Designed and implemented internal **GitLab CI/CD Pipelines** to automate build, test, and deployment workflows.
- Established a Unified Frontend Framework with **Next.js**, enhancing development efficiency and code reuse.

FULL-STACK ENGINEER / TECHNICAL PROJECT MANAGER, BUSINESS UNIT

Jul. 2018 – Jun. 2022

- Led and delivered over **45 web and enterprise projects**, including **E-Commerce**, **POS**, and **GIS-Based Systems**.
- Developed full-stack solutions using **ASP.NET Core**, **Laravel**, **Next.js**, and **Vue.js** with **MSSQL** and **MySQL** databases.
- Integrated multiple third-party and enterprise services such as **Payment Gateways**, **LINE OA**, and **POS Systems**.
- Refactored legacy **.NET Framework** backends into cross-platform, containerizable **.NET Core**.
- Supported clients on **Azure Cloud** deployment of web apps, VMs, and databases.

Internship

Taichung, Taiwan

SOFTWARE ENGINEER INTERN

Jul. 2015 – Jun. 2018

- Three successive internships at IT companies: **ISCOM Inc.**, **BitCloud Co.**, and **RCE Power Inc.**
- Web application development, front-end performance work, and internal data-collection automation.

Technical Skills

Cloud & Container Orchestration	Kubernetes (AKS, Talos Linux, Kubespray), Docker, Helm, Flux CD, Harbor, Karmada
Observability	Prometheus, Grafana, Loki, Tempo, Thanos, OpenTelemetry (OTLP), Alertmanager
AI & LLM Systems	LLM Agents, RAG, MCP, Prompt Engineering
Networking & Systems	Linux System Administration, IPv4 Networking, eBPF, MetallB, Traefik, Istio, Calico, Cilium
Infrastructure & Automation	Proxmox VE, Harvester, Terraform (OpenTofu), Ansible, Velero, GitLab CI/CD, GitOps
Security & Identity	Keycloak, HashiCorp Vault (OpenBao), NetworkPolicy, Cert-Manager, OAuth2 / OIDC
Programming & Scripting	Bash, Python, Go, C++, C# (.NET MVC/Core), PHP (Laravel), JavaScript (Vue.js / Next.js), SQL (MSSQL / MySQL)

Publications

An O-RAN Compliant Intent-Based MANO Framework for End-to-End Network Slicing

YI-HONG LIN*, CHIEN-HSIEN CHU, WEN-JU CHIANG, AND CHIEN-CHAO TSENG

- BIOTC 2025 (oral), IIHMSP 2025 (poster).

Invited Talks

Dec. 2025	Guest Speaker , Advanced Kubernetes Networking and Troubleshooting	NYCU
Sep. 2024	Guest Speaker , Introduction to Kubernetes and Its Networking Fundamentals	NYCU
Mar. 2022	Invited Lecturer , Exploring Trends and Career Paths in the Information Industry	CSMU

Awards & Honors

Academic Excellence Award , National Yang Ming Chiao Tung University (NYCU)	Spring 2024
Academic Excellence Award , National Yang Ming Chiao Tung University (NYCU)	Fall 2023